

APM1220: Applied Multivariate Data Analysis

FA 2

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Problem 3.1

Given the data matrix $X = \begin{bmatrix} 9 & 1 \\ 5 & 3 \\ 1 & 2 \end{bmatrix}$.

(a) Scatter plot in $p = 2$ dimensions

The sample mean is $\bar{x} = \begin{bmatrix} (9+5+1)/3 \\ (1+3+2)/3 \end{bmatrix} = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$. (On a 2D Cartesian plane, the points are located at $(9, 1)$, $(5, 3)$, and $(1, 2)$, with the sample mean plotted exactly in the center at $(5, 2)$.)

(b) $n = 3$ dimensional representation

The deviation vectors are calculated as $y_i - \bar{x}_i \mathbf{1}$:

$$d_1 = y_1 - \bar{x}_1 \mathbf{1} = \begin{bmatrix} 9 \\ 5 \\ 1 \end{bmatrix} - \begin{bmatrix} 5 \\ 5 \\ 5 \end{bmatrix} = \begin{bmatrix} 4 \\ 0 \\ -4 \end{bmatrix}$$
$$d_2 = y_2 - \bar{x}_2 \mathbf{1} = \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix} - \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$

(c) Lengths, Cosine, and Relationships

Length of d_1 : $L_1 = \sqrt{4^2 + 0^2 + (-4)^2} = \sqrt{32} = 4\sqrt{2}$.

Length of d_2 : $L_2 = \sqrt{(-1)^2 + 1^2 + 0^2} = \sqrt{2}$.

Cosine of the angle θ :

$$\cos(\theta) = \frac{d_1 \cdot d_2}{L_1 L_2} = \frac{4(-1) + 0(1) + (-4)(0)}{\sqrt{32}\sqrt{2}} = \frac{-4}{8} = -0.5$$

Relation to S_n and R : The squared lengths of the deviation vectors relate to the sample variances via $L_1^2 = ns_{11}$ and $L_2^2 = ns_{22}$. The cosine of the angle between them is exactly the sample correlation coefficient, $r_{12} = -0.5$.

Problem 3.6

Given $X = \begin{bmatrix} -1 & 3 & -2 \\ 2 & 4 & 2 \\ 5 & 2 & 3 \end{bmatrix}$. The mean vector is $\bar{x} = [2, 3, 1]'$.

(a) Matrix of deviations

$$X_d = X - \mathbf{1}\bar{x}' = \begin{bmatrix} -1-2 & 3-3 & -2-1 \\ 2-2 & 4-3 & 2-1 \\ 5-2 & 2-3 & 3-1 \end{bmatrix} = \begin{bmatrix} -3 & 0 & -3 \\ 0 & 1 & 1 \\ 3 & -1 & 2 \end{bmatrix}$$

This matrix is **not** of full rank. The sum of the rows in a mean-corrected deviation matrix is always the zero vector, which means the rows are linearly dependent. The maximum rank here is $n - 1 = 2$.

(b) S and Generalized Sample Variance

$$S = \frac{1}{n-1} X_d' X_d = \frac{1}{2} \begin{bmatrix} -3 & 0 & 3 \\ 0 & 1 & -1 \\ -3 & 1 & 2 \end{bmatrix} \begin{bmatrix} -3 & 0 & -3 \\ 0 & 1 & 1 \\ 3 & -1 & 2 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 18 & -3 & 15 \\ -3 & 2 & -1 \\ 15 & -1 & 14 \end{bmatrix} = \begin{bmatrix} 9 & -1.5 & 7.5 \\ -1.5 & 1 & -0.5 \\ 7.5 & -0.5 & 7 \end{bmatrix}$$

The generalized sample variance is $|S|$. Since the maximum rank of X_d is 2 (less than $p = 3$), $|S| = 0$. Geometrically, this means the 3-dimensional volume spanned by the deviation vectors is zero because they are entirely restricted to a 2D plane.

(c) Total Sample Variance

The total sample variance is the trace of S :

$$\text{Total Variance} = \text{tr}(S) = s_{11} + s_{22} + s_{33} = 9 + 1 + 7 = 17$$

Problem 3.9

(a) Mean corrected data matrix

The mean vector is $\bar{x} = [16, 18, 34]'$.

$$X_d = \begin{bmatrix} 12-16 & 17-18 & 29-34 \\ 18-16 & 20-18 & 30-34 \\ 14-16 & 16-18 & 30-34 \\ 20-16 & 18-18 & 35-34 \\ 16-16 & 19-18 & 35-34 \end{bmatrix} = \begin{bmatrix} -4 & -1 & -5 \\ 2 & 2 & 4 \\ -2 & -2 & -4 \\ 4 & 0 & 4 \\ 0 & 1 & 1 \end{bmatrix}$$

The columns are linearly dependent. Let $a' = [1, 1, -1]$. Multiplying X_d by a yields the zero vector, confirming the linear dependence.

(b) Sample Covariance Matrix S

$$S = \frac{1}{4}X_d'X_d = \frac{1}{4} \begin{bmatrix} -4 & 2 & -2 & 4 & 0 \\ -1 & 2 & -2 & 0 & 1 \\ -5 & 4 & -4 & 4 & 1 \end{bmatrix} \begin{bmatrix} -4 & -1 & -5 \\ 2 & 2 & 4 \\ -2 & -2 & -4 \\ 4 & 0 & 4 \\ 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 10 & 3 & 13 \\ 3 & 2.5 & 5.5 \\ 13 & 5.5 & 18.5 \end{bmatrix}$$

The generalized variance $|S| = 0$ because the columns are dependent. Verifying $Sa = 0$:

$$S \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 10(1) + 3(1) - 13(1) \\ 3(1) + 2.5(1) - 5.5(1) \\ 13(1) + 5.5(1) - 18.5(1) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

(c) Verify third column

Looking at the original matrix X , the elements of the third column equal the sum of the first two: $12 + 17 = 29$, $18 + 20 = 38$, $14 + 16 = 30$, $20 + 18 = 38$, $16 + 19 = 35$. Thus, the linear combination $x_1 + x_2 - x_3 = 0$ holds, making $a = [1, 1, -1]'$.

Problem 3.18

Let total energy $T = x_1 + x_2 + x_3 + x_4 = \mathbf{1}'x$.

(a) Total energy consumption statistics

Sample Mean: $\bar{T} = \mathbf{1}'\bar{x} = 0.766 + 0.508 + 0.438 + 0.161 = 1.873$.

Sample Variance: $S_T = \mathbf{1}'S\mathbf{1} = \text{Sum of all elements in } S = 1.76 + 1.398 + 0.511 + 0.245 = 3.914$.

(b) Excess of petroleum over natural gas

Let $E = x_1 - x_2 = a'x$ where $a = [1, -1, 0, 0]'$.

Sample Mean: $\bar{E} = a'\bar{x} = 0.766 - 0.508 = 0.258$.

Sample Variance: $Var(E) = a'Sa = s_{11} + s_{22} - 2s_{12} = 0.856 + 0.568 - 2(0.635) = 0.154$.

Sample Covariance with total: $Cov(E, T) = a'S\mathbf{1} = (\text{sum of row 1}) - (\text{sum of row 2}) = 1.760 - 1.398 = 0.362$.

Problem 3.20

(a) Summary Statistics Method

Computing summary statistics from the 25 observations: Mean of x_1 : $\bar{x}_1 = 11.716$

Mean of x_2 : $\bar{x}_2 = 16.976$

Sample variance of x_1 : $s_{11} \approx 36.810$

Sample variance of x_2 : $s_{22} \approx 75.712$

Sample covariance: $s_{12} \approx -12.937$

Sample mean of difference: $\bar{d} = \bar{x}_2 - \bar{x}_1 = 16.976 - 11.716 = 5.260$

Variance of difference: $s_{dd} = s_{11} + s_{22} - 2s_{12} = 36.810 + 75.712 - 2(-12.937) \approx 138.39$

(b) Individual Values Method

Calculating the individual differences $d_j = x_{j2} - x_{j1}$ for all 25 rows (e.g., $13.7 - 12.5 = 1.2$, $16.5 - 14.5 = 2.0$, etc.). Sum of $d_j = 131.5 \implies$ Mean $\bar{d} = \frac{131.5}{25} = 5.260$.

Sum of squares of $d_j = \sum d_j^2 = 4013.03$.

Sample Variance = $\frac{1}{24} (4013.03 - 25(5.260)^2) = \frac{3321.34}{24} \approx 138.39$.

As expected, calculating the variance from individual differences produces the exact same result as using the summary statistics from part a.